When Does Aggregating Explanations Work?
OrganSMNIST Results

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This report consolidates the released `OrganSMNIST` experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, NOISE-S, NOISE-K

Result source snapshot: `d6e4b9bbfb3c` from Workshop-When-Does-Explanations.

Table 1: NAIVE results for `OrganSMNIST` with `ViT-B/16` (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized. The test split contains $n = 8827$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Best Individual	0.057	0.347	0.203	0.499	0.046	-0.244	-0.001	-0.300	0.036	-0.237	-0.014	-0.269	0.011	-0.089	-0.035	-0.128	0.066	-0.029	0.077	-0.024
Simple Averaging	0.048	0.383	0.162	0.546	0.018	-0.219	0.020	-0.264	0.021	-0.217	0.007	-0.239	-0.006	-0.114	-0.012	-0.140	0.031	-0.064	0.029	-0.058
Borda	0.043	0.385	0.147	0.543	0.025	-0.223	0.028	-0.272	0.025	-0.222	0.017	-0.249	-0.001	-0.107	-0.004	-0.139	0.036	-0.050	0.048	-0.045
Kemeny–Young	0.044	0.396	0.152	0.552	0.027	-0.206	0.030	-0.259	0.026	-0.207	0.019	-0.236	-0.002	-0.094	-0.011	-0.129	0.029	-0.044	0.039	-0.042
RRF	0.040	0.393	0.152	0.550	0.034	-0.207	0.030	-0.258	0.029	-0.208	0.020	-0.235	0.001	-0.088	-0.007	-0.121	0.032	-0.047	0.039	-0.046
Schulze	0.043	0.387	0.150	0.545	0.028	-0.219	0.029	-0.269	0.028	-0.219	0.019	-0.248	0.001	-0.101	-0.005	-0.130	0.032	-0.049	0.043	-0.046

Table 2: NOISE-S results for `OrganSMNIST` with `ViT-B/16` (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	2	0.053	0.348	0.191	0.502	0.036	-0.249	-0.005	-0.305	0.033	-0.245	-0.007	-0.275	0.011	-0.121	-0.032	-0.158	0.063	-0.039	0.073	-0.038
Borda	2	0.057	0.350	0.192	0.507	0.028	-0.250	-0.009	-0.301	0.027	-0.247	-0.012	-0.275	0.008	-0.120	-0.033	-0.154	0.067	-0.035	0.076	-0.031
Kemeny–Young	2	0.057	0.350	0.192	0.507	0.028	-0.250	-0.009	-0.301	0.027	-0.247	-0.012	-0.275	0.008	-0.120	-0.033	-0.154	0.067	-0.035	0.076	-0.031
RRF	2	0.057	0.348	0.191	0.504	0.030	-0.252	-0.006	-0.303	0.028	-0.248	-0.011	-0.274	0.010	-0.121	-0.033	-0.156	0.066	-0.037	0.078	-0.032
Schulze	2	0.057	0.350	0.192	0.507	0.028	-0.250	-0.009	-0.301	0.027	-0.247	-0.012	-0.275	0.008	-0.120	-0.033	-0.154	0.067	-0.035	0.076	-0.031

Table 3: NOISE-K results for `OrganSMNIST` with `ViT-B/16` (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_{\mathtt{g}} = 0.15$, $\pi_{\mathtt{p}} = 0.05$, $\sigma_{\mathtt{s}} = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all `R` columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		$\mathtt{F}\uparrow$	$\bar{\mathtt{F}}\downarrow$	$\mathtt{C}\uparrow$	$\bar{\mathtt{C}}\downarrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{g}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{p}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{s}}_{\bar{\mathtt{C}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{F}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{F}}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\mathtt{C}}\uparrow$	$\mathtt{R}^{\mathtt{a}}_{\bar{\mathtt{C}}}\uparrow$
Simple Averaging	3	0.054	0.362	0.186	0.519	0.030	-0.233	0.012	-0.284	0.029	-0.227	0.002	-0.253	-0.002	-0.111	-0.027	-0.144	0.041	-0.060	0.031	-0.058
Borda	3	0.056	0.348	0.189	0.506	0.031	-0.246	0.006	-0.296	0.032	-0.244	0.000	-0.270	0.003	-0.118	-0.029	-0.151	0.068	-0.035	0.076	-0.030
Kemeny–Young	3	0.059	0.344	0.203	0.500	0.039	-0.251	-0.003	-0.303	0.036	-0.246	-0.011	-0.274	0.008	-0.114	-0.037	-0.148	0.062	-0.040	0.069	-0.033
RRF	3	0.060	0.348	0.198	0.507	0.033	-0.243	0.003	-0.291	0.033	-0.237	-0.006	-0.262	0.004	-0.114	-0.030	-0.145	0.061	-0.036	0.068	-0.027
Schulze	3	0.059	0.346	0.194	0.504	0.032	-0.248	0.006	-0.298	0.032	-0.246	-0.003	-0.271	0.003	-0.119	-0.030	-0.150	0.066	-0.033	0.074	-0.026